

**Problem 1**

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**Problem 1** Determine whether the following statements are true or false and give either an explanation or a counterexample.

- (a) The function  $f(x) = \sqrt{x}$  has a local maximum on the interval  $[0, 1]$ .
- (b) If  $f'(2) = 0$ , then  $x = 2$  is either a local maximum or local minimum of  $f$ .